

## Akka vs. CrewAI

A comparison for teams building agentic AI

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**CrewAI gets you to a multi-agent demo fast; with Akka the production system is owned by the platform, not by you.**

CrewAI is an ergonomic Python framework for prototyping role-based crews and event-driven flows. It does not provide a durable execution runtime, active-active HA/DR, a numeric availability SLA, or embedded governance — those you build, integrate, and operate yourself. Akka delivers them as one runtime, under contract.

**None**

CREWAI PUBLISHED SLA

**99.9999%**

AKKA SLA

**4ms**

AKKA STATE READS

**5B**

TOKENS / SEC BENCHMARK

| DIMENSION              | CREWAI                                                                                                                                       | AKKA                                                                                                              |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| What it is             | A Python multi-agent framework (Crews + Flows), plus the AMP control plane                                                                   | A full-stack agentic systems platform                                                                             |
| Scope                  | Agent authoring and orchestration; durable runtime, HA/DR, streaming, and governance are sourced, integrated, and operated by the customer   | Orchestration, agents, memory, streaming, APIs, observability, and governance on one runtime                      |
| Durable execution      | @persist() saves flow state; crash recovery is not built in — failures detected and replayed manually; the inner agent loop is not persisted | Event-sourced durable state; runtime handles recovery automatically                                               |
| Availability SLA       | None published                                                                                                                               | 99.9999% — entire platform, backed by indemnities                                                                 |
| HA / DR                | No published active-active or RTO/RPO posture                                                                                                | Active-active across regions; sub-1-minute RTO, zero-byte RPO                                                     |
| State / memory         | Configurable memory; flow state persisted to a configured backend                                                                            | Durable in-memory, 4ms reads / sub-10ms writes, PB cache, replayable                                              |
| Real-time streaming    | Not part of the framework                                                                                                                    | Built into the runtime — backpressured, petabyte-scale, no external broker                                        |
| Governance / EU AI Act | Guardrails, tracing, audit logging; no inline EU AI Act enforcement, classification, or sealed artifact                                      | Aspect-woven runtime enforcement + full pre-production governance                                                 |
| Cost model             | Execution-metered (Enterprise: up to 30,000/mo, then \$0.50 each) + separate infra for the rest                                              | Shared compute; up to 90% lower infrastructure for the same workload                                              |
| Certifications         | SOC 2, HIPAA, GDPR                                                                                                                           | 19 standards (SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF) |

### 1. CrewAI Is a Framework; Akka Is a Platform

CrewAI is a Python framework for building multi-agent systems — Crews of role-based autonomous agents and event-driven Flows — built from scratch, [independent of LangChain](#). It is fast and ergonomic to a working multi-agent demo, and that is its genuine strength. Production is a different problem: with CrewAI, the durable runtime, high availability, streaming, and governance around the agents are sourced, integrated, and operated by the customer.

CrewAI Enterprise / AMP (the [Agent Management Platform](#)) adds a control plane on top — a visual editor, an AI copilot, deployment to managed cloud or a private VPC, RBAC, SSO, tracing, and task guardrails. AMP makes the framework easier to operate. It does not change what the framework is: an agent-authoring layer, not a guaranteed runtime.

| Capability                                  | CrewAI                              | Akka                                      |
|---------------------------------------------|-------------------------------------|-------------------------------------------|
| Multi-agent authoring (crews, roles, flows) | Yes                                 | Yes                                       |
| Durable execution runtime                   | Manual (@persist() + manual replay) | Built in, automatic                       |
| Active-active HA / DR                       | No published posture                | Built in                                  |
| Durable memory                              | Configurable, backend-dependent     | Built in, 4ms / sub-10ms                  |
| Real-time streaming                         | Not in the framework                | Built in, backpressured, petabyte-scale   |
| Governance / policy enforcement             | Guardrails + tracing only           | Inline, runtime-embedded                  |
| Pre-production governance                   | None                                | Classification, sign-offs, sealed posture |

## 2. Reliability: Durable Execution, Availability, and Disaster Recovery

CrewAI does not publish a numeric availability SLA, an active-active HA/DR posture, or RTO/RPO targets. Durability is partial and manual: the [@persist\(\) decorator](#) saves flow state at chosen points, but crash recovery is not built in — the team detects the failure and triggers the replay themselves, and the autonomous agent reasoning loop is not persisted, so a crash inside it cannot be resumed. [Diagrid](#) sells a Dapr-based product specifically to add "true durable execution" to CrewAI, which confirms the gap from the ecosystem's own side.

| Metric                  | CrewAI                                                  | Akka                         |
|-------------------------|---------------------------------------------------------|------------------------------|
| Availability SLA        | None published                                          | 99.9999%                     |
| Allowed downtime / year | Not specified                                           | ~31 seconds                  |
| Crash recovery          | Manual detect + manual replay; inner loop not persisted | Automatic, runtime-handled   |
| RTO                     | No published target                                     | Sub-1 minute                 |
| RPO                     | No published target                                     | Zero byte                    |
| HA / DR                 | No published active-active posture                      | Active-active across regions |
| SLA scope               | The customer owns application availability              | The entire platform          |

Akka's reliability is structural: actor-based concurrency and clustering, durable sharded in-memory state replayable from its event journal, and brokerless backpressured streaming with a 1-minute RTO and 0-byte RPO. The platform owns the SLA, with 24/7 SRE and contractual indemnities.

### 3. Up to 90% Cheaper to Operate

AI systems built with Akka are up to **90% cheaper to operate** than Python-based systems — a function of the infrastructure required for the same agentic transaction volume, not list price. The drivers are actor concurrency, shared compute, and micro-checkpointing: Akka processes ~10 trillion tokens per core per year versus ~2 trillion for comparable solutions, using ~80% less compute than Python-based frameworks. Manulife reported up to 300% more concurrency and 30–50% faster processing after porting from Python.

### 4. Governance and the EU AI Act

CrewAI provides task guardrails, tracing, audit logging, and RBAC, and holds SOC 2, HIPAA, and GDPR — security and operational controls. It does not provide inline EU AI Act enforcement: no real-time policy enforcement woven into the runtime, no decision explainability, no human pause/override of a running process, no hash-chained immutable interaction ledger, no pre-deployment classification, and no sealed audit artifact. Comprehensive EU AI Act conformance on CrewAI requires integrating a separate governance platform.

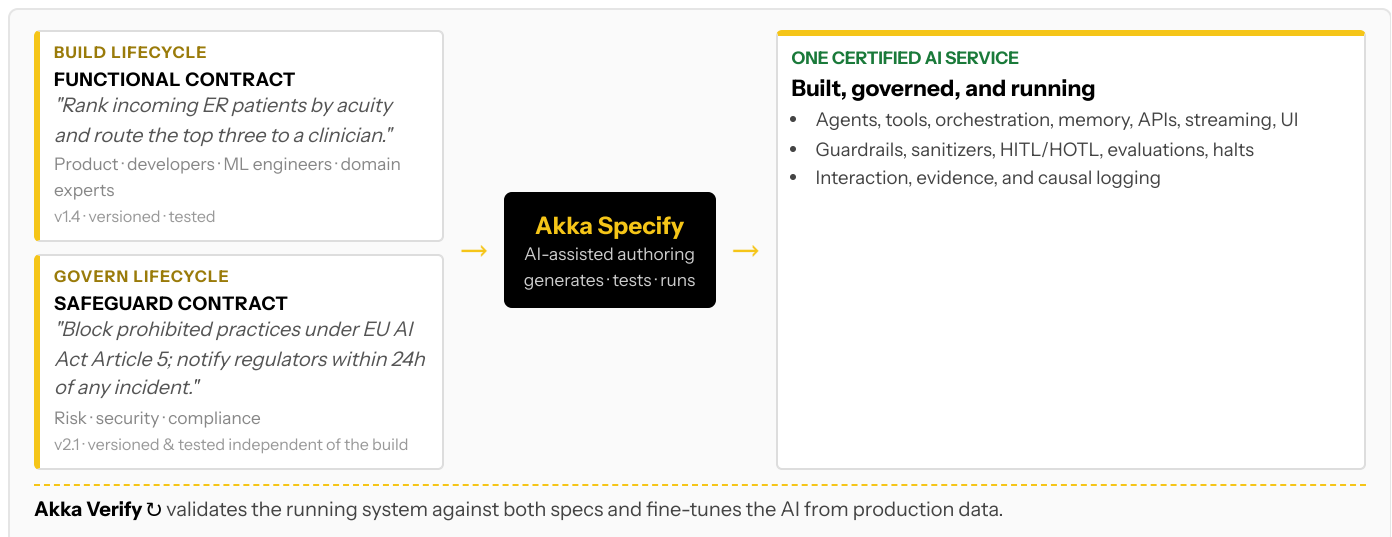
#### The penalties are enforceable now

| Violation                            | Maximum Fine                            |
|--------------------------------------|-----------------------------------------|
| Prohibited AI practices (Art. 5)     | <b>EUR 35M or 7% global turnover</b>    |
| High-risk obligations (Art. 9-15)    | <b>EUR 15M or 3% global turnover</b>    |
| Incorrect information to authorities | <b>EUR 7.5M or 1.5% global turnover</b> |

High-risk AI carries a 10-year logging-retention obligation (Art. 72).

### 5. Two Lifecycles, One Certified System

Building on CrewAI means engineers writing Python crews and flows; there is no built-in path for a risk officer or compliance owner to author and version a governance contract independently of the application, and no governance lifecycle in the framework. Akka runs two independent lifecycles on one platform via **Akka Specify**:



The build lifecycle and the governance lifecycle are versioned and tested independently, by different audiences — an audience and a workflow CrewAI has no equivalent for.

### The Execution Meter

CrewAI is a Python framework billed by execution — the Enterprise tier includes up to 30,000 executions per month, with additional executions at \$0.50 each — and that meter covers agent runs only. The durable runtime, HA/DR, streaming, observability, and governance are separate infrastructure the customer provisions and pays for. Akka runs all of it on one shared-compute runtime for a fixed annual fee finance can forecast, rather than per-execution metering that moves with load.

## 6. Real-Time Streaming at Petabyte Scale

CrewAI has no streaming engine; real-time pipelines are provisioned separately. Akka's streaming is built into the runtime — continuous, backpressured, **petabyte-scale, in-memory**, with no external broker — powering both agent feedback loops and high-throughput data processing (the engine behind Tubi's real-time hyper-personalization at 5 billion tokens per second).

## 7. For the Buyer: Risk, Compliance, and Accountability

| Buyer concern                | CrewAI                                                                                                       | Akka                                                                                                                                                                           |
|------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Certifications & audits      | SOC 2, HIPAA, GDPR                                                                                           | 19 standards — SOC 2 II + public SOC 3, ISO 27001/42001, HIPAA, PCI DSS, GDPR, NIS2, DORA, EU AI Act, NIST AI RMF — plus annual pen tests, SBOMs, 40+ policies (trust.akka.io) |
| Scope of accountability      | Agent framework + control plane; you integrate and operate the durable runtime, HA/DR, and governance        | One platform, one SLA, 24/7 SRE — Akka owns the running system                                                                                                                 |
| Availability commitment      | No published numeric SLA                                                                                     | 99.9999%, backed by contractual indemnities                                                                                                                                    |
| Risk transfer                | Standard cloud terms                                                                                         | Availability and data-integrity guarantees backed by contractual indemnities                                                                                                   |
| Track record & funding model | Venture-funded: \$18M total, Series A led by Insight Partners (Oct 2024); used across many Fortune 500 firms | Profitable and self-funding; 18 years and 100,000+ deployments (52 banks); Dell Technologies Capital is largest shareholder, a customer, and an AI partner                     |
| Budget predictability        | Per-execution metering that scales with load                                                                 | Fixed annual fee finance can forecast                                                                                                                                          |

CrewAI has real enterprise traction — it reports use across a large share of the Fortune 500 and hundreds of millions of agent executions a month. The decision is not whether CrewAI works for prototyping; it is who owns production. With CrewAI the customer owns the durable runtime, the availability target, the DR plan, and the governance stack. With Akka the platform owns them, under contract.

## Customers Running Agentic and Real-Time Systems on Akka

|                                                                                            |                                                                          |                                                                             |                                                                                     |                                                                                     |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| <b>Manulife</b><br><b>2,000</b><br>developers across 100 projects on one governed platform | <b>Tubi</b><br><b>5B tok/s</b><br>real-time hyper-personalization engine | <b>Swiggy</b><br><b>71ms</b><br>order-assignment AI, ~50% latency reduction | <b>John Deere</b><br><b>1,000+</b><br>tractor sensors turned into real-time insight | <b>Verizon</b><br><b>750%</b><br>order-processing capacity gain; 6s → 2.4s response |
|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|

## Common Questions

### We already prototype with CrewAI. Why move to Akka?

CrewAI is good for getting a multi-agent system working quickly. The gap appears in production: durable execution is manual, there is no published availability SLA or active-active HA/DR, and EU AI Act governance is not enforced in the runtime. Akka provides all of that as one platform, so you do not assemble and operate it yourself. You can keep prototyping in CrewAI and standardize production on Akka.

### Doesn't CrewAI's @persist() give us durable execution?

@persist() saves flow state to a configured backend, but crash recovery is not automatic — your team detects the failure and triggers the replay, and the agent's inner reasoning loop is not persisted, so a crash there cannot be resumed. Third parties such as Diagrid sell durable-execution add-ons specifically to fill this gap. Akka's runtime recovers automatically from an event journal with a sub-1-minute RTO and zero-byte RPO.

### Can we add governance on top of CrewAI?

You can add guardrails, tracing, and audit logging, and integrate a separate governance platform. But the EU AI Act expects enforcement inline to the runtime: immutable records witnessed as they happen, human override of running processes, pre-deployment classification, and a sealed audit artifact. Bolt-on tools observe after the fact and cannot gate a deployment. Akka embeds all of this and covers pre-deployment governance.

### Is CrewAI cheaper because the framework is open source?

The core framework is open source, but production CrewAI means CrewAI Enterprise / AMP (execution-metered) or self-hosting — where you build and operate the durable runtime, HA/DR, streaming, observability, and governance yourself. Akka's shared-compute model is up to 90% cheaper to operate than the equivalent assembled stack, on a fixed annual fee.

## Sources

**CrewAI OSS framework:** [github.com/crewAIInc/crewAI](https://github.com/crewAIInc/crewAI) — Python framework for role-based crews and event-driven flows, built from scratch, independent of LangChain

**CrewAI AMP / Enterprise:** [docs.crewai.com/en/enterprise/introduction](https://docs.crewai.com/en/enterprise/introduction) — control plane: visual editor, AI copilot, cloud or private VPC deploy, RBAC, SSO (Entra/Okta), tracing, task guardrails

**CrewAI durable-execution gap:** [speakeasy.com/blog/ai-agent-framework-comparison](https://speakeasy.com/blog/ai-agent-framework-comparison); [diagrid.io/solutions/crewai-production](https://diagrid.io/solutions/crewai-production) — @persist() saves flow state; crash recovery not built in; manual detect + replay; inner loop not persisted; Diagrid adds true durable execution via Dapr

**CrewAI compliance:** [github.com/crewAIInc/crewAI/discussions/2404](https://github.com/crewAIInc/crewAI/discussions/2404) — SOC 2, HIPAA, GDPR, RBAC, audit logging (no published availability SLA, active-active HA/DR, or EU AI Act enforcement found)

**CrewAI pricing:** [techjacksolutions.com/ai-tools/crewai/crewai-pricing](https://techjacksolutions.com/ai-tools/crewai/crewai-pricing);

[toolworthy.ai/tool/crewai](https://toolworthy.ai/tool/crewai) — execution-metered; Enterprise up to 30,000 executions/mo, additional at \$0.50 each

**CrewAI funding & traction:** [siliconangle.com/2024/10/22](https://siliconangle.com/2024/10/22); [insightpartners.com](https://insightpartners.com) — \$18M total, Series A led by Insight Partners, Oct 2024; used across a large share of the Fortune 500; ~450M agents/month (Jan 2026)

**Akka trust center:** [trust.akka.io](https://trust.akka.io) — 19 compliance standards; SOC 2 II + public SOC 3; annual pen tests, SBOMs, 40+ policies

**Akka performance:** [akka.io/blog/go-slow-to-go-fast](https://akka.io/blog/go-slow-to-go-fast) — Manulife up to 300% more concurrency, 30–50% faster; ~10T vs ~2T tokens/core; ~80% less compute than Python

**Akka platform:** 99.9999% availability, active-active HA/DR, sub-1 min RTO, zero byte RPO (contractual indemnities); 186 regulations / 877 controls / 288 carrying a financial penalty; 100,000+ deployments / 18 years; profitable; Dell Technologies Capital